SmartSteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics

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Respiratory illnesses such as pneumonia, bronchiolitis, and asthma remain leading causes of morbidity and mortality in rural Philippine communities, where diagnostic resources are limited and frontline health workers often rely on manual auscultation and subjective clinical judgment. This study introduces SmartSteth, a convolutional neural network (CNN)-driven auscultation instrument designed to enhance early detection of lower respiratory tract diseases in low-resource settings. The device integrates a MEMS microphone with an ESP32 microcontroller and a web-based AI interface to analyze lung sounds in real time, providing immediate, interpretable diagnostic feedback.

A labeled dataset of respiratory sounds—including normal, asthmatic, pneumonic, and bronchiolitic recordings—was collected using digital stethoscopes and processed into Mel-spectrograms. These were used to train a CNN model via Edge Impulse, optimized through iterative validation to ensure reliable classification. The model was deployed within a web application that receives audio input from the device, performs preprocessing, and displays classification results. Hardware assembly, software integration, and usability testing were conducted to evaluate diagnostic accuracy, portability, and user experience.

SmartSteth demonstrated strong potential for improving clinical decision-making, reducing diagnostic delays, and enhancing patient-provider communication. Its offline functionality, intuitive interface, and modular design support deployment in barangay health stations, mobile clinics, and home visits. The device aligns with global health priorities by offering a scalable, cost-effective solution for respiratory screening and surveillance. By bridging technological innovation with community-based care, SmartSteth contributes to equitable healthcare access and supports key United Nations Sustainable Development Goals.

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